

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 13: Section and Mid-Point Formula

Every formula and result of the chapter on one sheet

1	Section formula ($m_1 : m_2$)	$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$
2	Section formula ($k : 1$)	$\left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1} \right)$
3	Mid-point formula	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
4	Centroid of a triangle	$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$
5	Centroid on a median	Divides it in the ratio 2 : 1 from the vertex
6	Points of trisection	Ratios 1 : 2 and 2 : 1
7	Point on the x-axis	Ordinate = 0
8	Point on the y-axis	Abcissa = 0
9	Parallelogram	The diagonals have the same mid-point
10	Centre of a circle	Mid-point of any diameter